SLAMs: Stellarators Linking Axisymmetric Mirrors

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Abstract. We present a computational design study of a stellarator–mirror hybrid concept, a Stellarator Linked Axisymmetric Mirror or SLAM, in which an optimised poloidally omnigenous (OP or QI) stellarator serves as the rotational-transform-generating element that links long, axisymmetric (circular cross-section) magnetic-mirrors. Starting from OP precursor configurations drawn from the DESC omnigenity database, we construct hybrid coil sets for field periods $n_{\mathrm{FP}} = 2, 3, 4$ SLAMs by separating the OP modular coils along the device midplane and inserting axisymmetric planar mirror coils into the resulting gap. Field-line tracing shows that nested vacuum flux surfaces survive this extrusion across a range of mirror-section lengths for all three field-periods, with the cross-section qualitatively preserved at the point where the mirror is inserted. We use the field line tracing data to map the shape of a surface and use the GVEC fixed boundary solver to calculate an ideal magnetohyrodynamic (MHD) equilibrium and evaluate physics metrics such as the neoclassical transport proxy ($\epsilon_{\mathrm{eff}}^{3/2}$) and the vacuum magnetic well ($V^{\dagger\dagger}$) for different mirror to stellarator ratios L/D, doing a first-principles analysis of such concepts using modern stellarator analysis tools.

Keywords: stellartor-mirror hybrids, stellarator optimization, stability, transport

1. Introduction

Magnetic-mirror and stellarator schemes sit at opposite ends of the magnetic-confinement design space. Axisymmetric mirrors are simple to build, achieve high local β in their low-field central cell, and have recently demonstrated high-field operation at 17 T using HTS magnets [1], with the engineering path to a break-even axisymmetric mirror device set out by Forest *et al* [2]. Their two persistent physics limitations are the open loss cone, which drives parallel particle losses despite multiple-mirror plugging [3], helical-mirror flow drive [4, 5], and edge-bias rotation [6], and the MHD interchange instability of the unfavourably-curved central cell [7], which earlier devices

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controlled by minimum-B coil geometries or by providing favourable curvature beyond the mirror throat as in the gas-dynamic trap [8]. Current high-field experiments achieve stabilization some combination of finite-Larmor-radius effects, sheared edge rotation, and an energetic-ion population [1]. Optimised stellarators face the opposite problem: they close the loss cone by construction, can be designed against ideal-MHD instabilities, and admit configurations with vanishing bootstrap current and near-omnigenous drift confinement [9–11], at the price of non-planar coils, non-axisymmetric vacuum vessels, and out-of-plane mechanical loads that drive the capital cost per unit plasma volume well above that of an equivalent mirror. A device that combined the engineering simplicity of the mirror with the closed field lines and drift confinement of the stellarator would therefore be of practical interest; the question is whether such a device can be built without sacrificing the properties that justify modern stellarator design in the first place.

This question has been pursued in several lines of work since the 1960s, in parallel with the development of the theoretical framework for omnigenous fields. The notion of an omnigenous field, one in which the bounce-averaged radial drift of trapped particles vanishes, was formalised by Hall and McNamara [12] and Catto and Hazeltine [13], following earlier explicit constructions of closed isodynamic fields by Palumbo [14], non-symmetric omnigenous solutions of finite extent were given by Cary and Shasharina [15], and the poloidally omnigenous subfamily — omnigenous configurations with poloidally-closed contours of $|\boldsymbol{B}|$ was introduced by Gori *et al* [16].

In parallel, the engineering architecture for a stellarator–mirror hybrid was set out in the DRAKON proposal of Glagolev *et al* [17]: two long straight mirrors joined by curvilinear elements (CRELs) such that the bulk plasma resides in low-field axisymmetric cells while the loss cone is closed by toroidal connection. Glowienka [18] designed a set of toroidally connected magnetic mirrors as a part of the Elmo Bumpy Torus (EBT) experiment. Moiseenko *et al* [19] studied a stellarator–mirror hybrid for fission–fusion applications, realising a local magnetic mirror in the Uragan-2M $l = 2$ torsatron by switching off a toroidal-field coil, and concluded that the resulting mirror cell loses the symmetry favourable for trapped-particle confinement. Mazzucato [20] described a racetrack reactor in which two $200m$ magnetic mirrors of mirror ratio $M = 2$ are connected by semicircular sections carrying $l = 2$ helical windings that supply rotation and shear, projecting ~ 13 GW of fusion power at ITER-like density and temperature. Strickler *et al* [21] and Spong and Harris [22] designed quasi-poloidal stellarator configurations that turned out to be remarkably close to racetrack shape. Recently, Feng *et al* [23] proposed a related linked-mirror device in which rotational transform is generated by a finite angle 2Θ between two non-parallel straight mirror sections, a modern reinterpretation of the Figure-8 stellarator [24], and reported good single-particle and neoclassical confinement in vacuum, including for ripple-trapped orbits. The theoretical basis underlying these proposals is the observation by Shafranov *et al* [25] that the property responsible for good guiding-centre confinement in both poloidally-closed mirrors and modern stellarators is the same quasi-symmetry, or, more

generally, omnigenity, so that optimised mirrors and poloidally omnigenous stellarators can in principle be joined without disrupting drift confinement. We refer to the resulting class of configurations as Stellarators Linking Axisymmetric Mirrors, or SLAMs.

This paper is organised as follows: in section 3 we introduce the fundamentals of an ideal MHD equilibrium, how it relates to physics metrics, and briefly introduce the DESC [26–28] and GVEC [29] codes. This paper provides SLAM-A as more of an initialization technique

2. Cost-benefit analysis of pursuing the SLAM

The engineering case for pursuing SLAM rests on a volumetric scaling argument that we set out here as motivation rather than as a quantitative cost model. Consider two devices of equal total plasma volume: a conventional QI stellarator of volume V_s^0, and a SLAM device consisting of a stellarator linker of volume V_s joined to an axisymmetric mirror cell of volume V_m, with $V_s + V_m = V_s^0$. Treating the capital cost of each section as proportional to its enclosed plasma volume, a zeroth-order approximation justified by the observation that the dominant cost drivers (magnets, vacuum vessel, support structure, blanket) all scale roughly with surface area or enclosed volume, we write the total costs as $C_s^0 = k_s V_s^0$ for the conventional stellarator and $C_{\mathrm{SLAM}} = k_s' V_s + k_m V_m$ for the hybrid, where k_s is the cost per unit plasma volume of a free-standing stellarator section, k_m is the corresponding quantity for an axisymmetric mirror section, and k_s' is the cost per unit volume of the stellarator section *within* a SLAM device, which may differ from k_s if, for example, neutron-flux localisation in the mirror cell relaxes the shielding requirement and permits modular coils to be brought closer to the plasma. Three dimensionless ratios then capture the essential dependence:

$$\alpha \equiv \frac{k_s}{k_m} \geq 1, \qquad \gamma \equiv \frac{k_s'}{k_s} \leq 1, \qquad V_r \equiv \frac{V_m}{V_s}, \tag{1}$$

where α measures the cost-per-volume penalty of three-dimensional stellarator engineering relative to axisymmetric mirror engineering, γ is the cost-reduction factor available to the stellarator section once it is embedded in a SLAM, and V_r is the design choice. In these variables, the ratio of conventional to SLAM cost at equal plasma volume reduces to

$$A \equiv \frac{C_s^0}{C_{\mathrm{SLAM}}} = \frac{1 + V_r}{\gamma + V_r/\alpha}. \tag{2}$$

The structure of A is transparent in two limits: $V_r \to 0$ gives $A \to 1/\gamma$, the saving from neutron-flux relaxation alone, while $V_r \to \infty$ gives $A \to \alpha/\gamma$, the full per-volume saving. Cost advantage $A > 1$ therefore requires either $\alpha > 1$ (mirror sections are intrinsically cheaper per cubic metre of plasma) or $\gamma < 1$ (the stellarator section benefits from neutron-flux localisation), and is amplified by increasing V_r until the asymptotic limit α/γ is approached. Two further relations link this scaling to plasma physics. Conservation of toroidal flux along a flux tube, $BA_\perp = \mathrm{const}$, gives $V_r = M_r L_r$, where $M_r \equiv B_s/B_m$ is the mirror ratio between the stellarator and mirror sections and L_r is

the ratio of mirror-section axial length to stellarator-section circumferential length; and the local plasma $\beta = 2\mu_0 p/B^2$ at fixed flux-surface pressure p scales as $\beta_m = M_r^2 \beta_s$, so that operating at large mirror ratio simultaneously raises plasma volume and magnetic-utilisation efficiency. We emphasise that the equations above are scaling estimates, not engineering cost models—values of α, γ, V_r and M_r in the favourable regime would all need to be substantiated by a full engineering analysis—but they identify the parameter range in which a stellarator–mirror hybrid is worth pursuing and they expose three operating tiers: a baseline tier with $V_r > 0$ at the natural QI mirror minimum and M_r of order unity (SLAM-A), a higher-magnetic-utilisation tier with $M_r \gg 1$ and isotropic pressure (SLAM-B), and a tier with $M_r \gg 1$ and anisotropic pressure $p_\perp \neq p_\parallel$ that localises fusion power density to the mirror cell (SLAM-C). The present paper addresses SLAM-A as the necessary first step.

3. Ideal MHD equilibrium

In this section, we briefly explain how we define and calculate an ideal MHD equilibrium in a SLAM.

A divergence-free magnetic field $\boldsymbol{B}$ can be written in the Clebsch form [30]

$$\boldsymbol{B} = \nabla\psi \times \nabla\alpha, \tag{3}$$

We will focus on solutions whose magnetic field lines lie on closed nested toroidal surfaces, known as flux surfaces. We label these surfaces using the enclosed toroidal flux ψ. On each flux surface, the lines of constant α coincide with the magnetic field lines. Thus, α is known as the field line label. We define $\alpha = \theta_{\text{PEST}} - \iota(\zeta - \zeta_0)$, where θ is the PEST straight field line angle, ζ is the generalized cylindrical toroidal angle, ζ_0 is a constant, and

$$\iota = \frac{\boldsymbol{B} \cdot \nabla\theta_{\text{PEST}}}{\boldsymbol{B} \cdot \nabla\zeta}, \tag{4}$$

is the pitch of the magnetic field lines on a flux surface, known as the rotational transform. Using the Clebsch form of the magnetic field, we solve the ideal MHD force balance equation

$$\boldsymbol{j} \times \boldsymbol{B} = \nabla p, \tag{5}$$

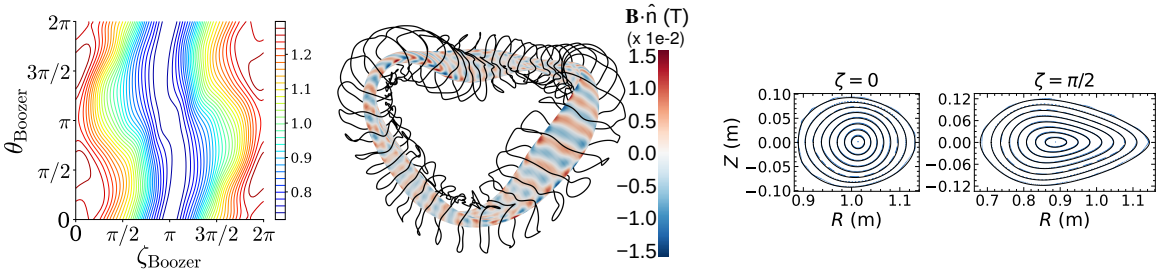
where the plasma current $\boldsymbol{j} = (\nabla \times \boldsymbol{B})/\mu_0$ from Ampere's law, p is the plasma pressure and μ_0 is the vacuum magnetic permeability. Unlike an axisymmetric case, for stellarators, we have to solve (5) as an optimization problem. We achieve this with the `DESC` stellarator optimization suite. `DESC` can simultaneously solve an equilibrium while optimizing for multiple objectives such as MHD stability, quasisymmetry, and many more. In this paper, we use some of these metric to generate various stellarator equilibria. The details of the `DESC` package are briefly provided in these references [28, 31–33].

4. Creating a SLAM-A coilset with DESC

In this section, we will describe in detail how we used DESC to design coils for a SLAM-A equilibrium. Contrary to the usual process of first designing an equilibrium (stage-one optimization) and then coils (stage-two optimization), we make a SLAM equilibrium by starting with coilsets from a poloidally omnigenous (OP) taken from the DESC omnigenity database, and adding mirror between the stellarator coils to form a stellarator-mirror hybrid coilset. followed by tracing the boundary from a single field lines to map a flux surface created by the coils and which becomes an input to the equilibrium code GVEC. Then, we vary some of the mirror coils and perform Quadratic Flux Minimizing (QFM) surface optimization to make the mirror cross section is circular without affecting the stellarator region. Finally, we map the SLAM-A surface by tracing a field line again to get the boundary shape and calculate the equilibrium using GVEC.

4.1. Creating the stellartor mirror hybrid coilset

We take an optimized OP equilibrium and its coilset from the DESC omnigenity database. The equilibrium is generated using stage-one optimization where the boundary shape is varied to ensure favorable physics properties and the coilset is design using stage-two optimization, where the boundary is fixed and coils are generated to minimize the normal field error $\boldsymbol{B} \cdot \hat{\boldsymbol{n}}$ on the boundary. The coilset and properties of the equilibrium are presented in figure 1.



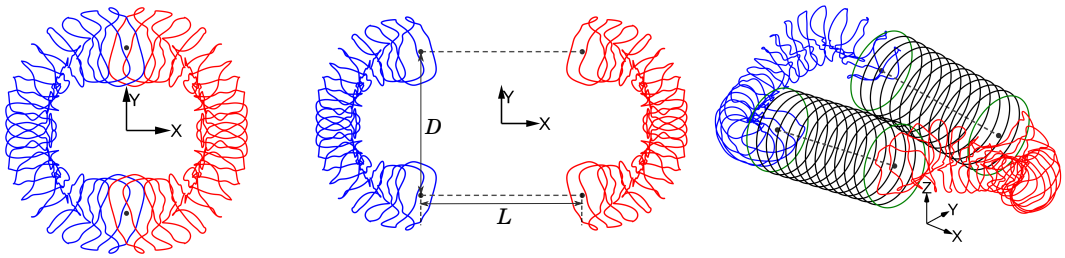
(a) $|\boldsymbol{B}|(\rho = 1.0)$ (b) $\boldsymbol{B}\cdot\hat{\boldsymbol{n}}(\rho = 1.0)$ and coils (c) Poincare trace

Figure 1: A $n_{\text{FP}} = 2$ vacuum OP equilibrium and its corresponding coilset taken from the DESC omnigenity database. In figure 1a, we represent the field strength in Boozer coordinates. In figure 1b, we present the OP coilset along with the normal magnetic field error $\boldsymbol{B} \cdot \hat{\boldsymbol{n}}$ on the equilibrium boundary. In figure 1c we trace the magnetic field lines from the coils and compare them with equilibrium flux surfaces. The coilset is able to reproduce the flux surfaces obtained from the fixed boundary DESC equilibrium.

To create a stellarator-mirror hybrid, we take the coils in the OP coilset, and separate them about the low-field region§, where the magnetic axis curvature vanishes.

§ One can separate the coils and add a mirror in a low-field and/or high-field region but adding a mirror in the low-field region is more advantageous as we discussed in section 2.

We then shift the coils and fill the gap with circular mirror coils, with all mirror coils having the same current, which is 50% higher than the current carried by the closest low-field stellarator coil. The process of converting a stellarator coilset to a stellarator-mirror hybrid coilset is demonstrated in figure 2.



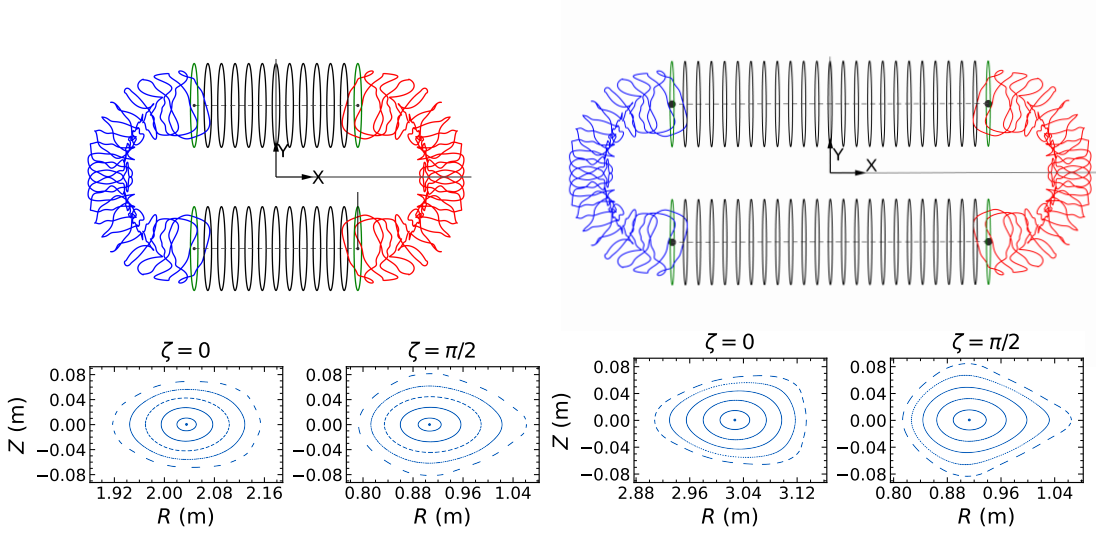
(a) Original (b) Separated along X axis (c) Stellarator-Mirror Hybrid

Figure 2: Process of making a stellarator-mirror hybrid coilset. We start with the original OP coilset in figure 2a, separate it at the point of lowest magnetic field strength and vanishing axis curvature, along the X axis by a distance L in figure 2, and using the line (dashed) joining the magnetic axis in the original coilset as the center, place circular mirror coils in the gap as shown in figure 2c.

We characterize every racetrack-shaped stellarator mirror hybrid using the ratio of the length of the mirror L to the major diameter of a stellarator $D = 2R$, L/D, where R is the major radius of the original OP stellarator. This parameter is related to L_r defined in section 2 such that $L_r = (1/\pi)L/D$. We now discuss two potential issues with the coil arrangement. First, the line joining the magnetic axis from the original coilset with the mirror coil center is non-smooth and can potentially change the physics properties of the stellarator-mirror hybrid. We will revisit this point when we analyze the physics properties in section number. Second, after adding the mirror coils, there is no guarantee that the magnetic field will form nested flux surfaces. To test this, we trace field lines from the Stellarator-Mirror hybrid coilset in figure 3.

We find that nested flux surfaces exist even after adding the circular mirror coils in the low-field region at least up to $L/D = 2$. In terms of the convention used by Feng *et al* [23], the angle between the two straight sections $2\Theta \neq 0$ for either case in figure 3. However, since the OP coilset stays the same for both the mirror lengths, $2\Theta(L/D = 2) < 2\Theta(L/D = 1)$, and the two mirrors become more parallel as we increase the mirror length, *i.e.*, $\lim_{L/D\to\infty} 2\Theta = 0$.

We also find that the nestedness of the stellarator-mirror hybrid is robust to the current in the mirror coils. Varying the current from $0.8 - 2.0$ of the current in the adjacent low-field coil sustains nestedness, but changes the flux surface shapes in both the stellarator the mirror regions. Furthermore, the low-field Poincaré trace strongly resembles the low-field Poincaré trace in the original OP coilset even after creating the hybrid. For further analysis, we choose the stellarator-mirror hybrid with $L/D = 1$.



(a) $L/D = 1.0$ (b) $L/D = 2.0$

Figure 3: Stellarator mirror hybrid coilset using the OP coilset presented in figure 1b with two different mirror lengths and their corresponding Poincaré plots on the planes where the black lines intersection the coils. Nested surfaces exist even after separating the OP coilset and adding mirror coils. Note that the cross-section in the mirror is not a circular (axisymmetric).

In the next section, we will use the existence of flux surfaces in a stellarator-mirror hybrid coilset to optimize the mirror section to become circular (axisymmetric). Need to clarify multiple times, especially in the beginning that "axisymmetric" in SLAM means circular.

4.2. Axisymmetrize the mirror region without changing the stellarator region

We take advantage of flux surface nestedness in the stellarator-mirror hybrid by finding a surface on which the normal magnetic field $\boldsymbol{B}\cdot\hat{\boldsymbol{n}}$ (or the quadratic flux $(\boldsymbol{B}\cdot\hat{\boldsymbol{n}})^2$). Such a surface is also knows as a Quadratic Flux Minimizing (QFM) surface. As the quadratic flux approaches zero on a QFM surface, it becomes a identical to a flux surface‖.

We will first create a QFM surface from the intial statellarator-mirror hybrid coilset shown in figure 3a, then use further optimization to axisymmetrize the mirror and create a SLAM while keeping the stellarator part of the SLAM unchanged. To find the QFM surface, we begin with a initial closed surface $\mathcal{S}_1$ parametrized using Fourier coefficients $R_{\mathrm{b},mn}, Z_{\mathrm{b},mn}$ and allow the coefficients to change while minimizing the quadratic flux $(\boldsymbol{B}\cdot\hat{\boldsymbol{n}})^2$. The QFM surface for the initial coilset is shown in figure ref

A QFM surface optimization is different from a conventional stage-one optimization, which optimizes the plasma boundary to ensure favorable physics properties, and stage-two optimization, which fixes the plasma boundary and modifies coil shapes to

‖ A QFM surface can also become an island surface which is why it is important to check existence of closed flux surfaces before performing QFM optimization.

become consistent with the fixed-boundary equilibrium. Here, we allow both the plasma boundary and coilset to change as long as the QFM surface corresponds to a flux surface that is consistent with the coilset without optimizing for physics metrics.

Using the QFMince the cross-section in the mirror region is non-circular (non-axisymmetric), w

$$\mathcal{F} = w_C f_{\text{QF}}^2 + w_D f^2 + w_E f_{\text{elongation}}^2 + w_O f_{\text{om}}^2, \tag{6}$$

where f_x are various objectives on the right side for omnigenity, distance from ideal-ballooning marginality, Mercier stability, boundary curvature, boundary elongation, and aspect ratio, and $w_A, w_B, w_C, w_D, w_E, w_O$ are weights used with each objective function. The exact definitions of these objectives are provided in **??**. At each iteration of a DESC optimization, $\mathcal{F}$ is minimized while satisfying (5). Mathematically,

$$\min(\mathcal{F}_{\text{stage-one}}), \quad \text{s.t.} \nabla \left(\mu_0 p + \frac{B^2}{2}\right) - \boldsymbol{B}\cdot\nabla\boldsymbol{B} = 0,\ \psi_{\text{b}} = \psi_{\text{b0}}, p = p_0(\psi), \iota = \iota_0(\psi) \tag{7}$$

To axisymmetrize the mirror region without changing the stellarator part, we perform a QFM surface optimization, where we allow some of the mirror coils to change until the mirror region becomes circular, while minimally changing the stellarator boundary.

QFM surf

solving a different stage-two optimization problem.

Adding the magnetic mirror region and axisymmetrizing changes the boundary and consequently the physics properties of the equilibrium. To understand how the physics properties are affected, we must use the new flux surface shape and solve a fixed boundary equilibrium. However, the race track and elongation axis shape Representation in Boozer coordinates and why that is a good coordinate system for a Important to underscore the inability of the QFM to do that. There is not iota information on a QFM! More robust than Guiliani's method for SLAMs.

4.3. Mapping flux surfaces by tracing a single field line with the FTGen algorithm

Adding the mirror section to an OP coilset and subsequently axisymmetrizing it changes the magnetic field and, consequently, the shape of the outermost nested flux surface. To calculate an ideal-MHD equilibrium for the resulting hybrid configuration, we therefore require a parametrization of this new boundary. Although the QFM optimization described in the previous section provides a geometric surface with small normal magnetic-field error, the QFM surface itself does not provide the rotational transform ι, poloidal current G or a straight-field-line parametrization. We instead obtain both directly from the magnetic field generated by the final coilset using a novel field-line-tracing surface-generation algorithm, which we refer to as **FTGen** (Flux Tube Generator).

The central idea of FTGen is that a sufficiently long trajectory of a single field line on an irrational flux surface samples the entire toroidal surface ¶. Rather than treating the trace as a one-dimensional curve, FTGen uses the rotational transform to assign successive toroidal transits of the field line different field-line labels. Consider a field line traced through an unwrapped, uniformly-spaced geometric toroidal angle $\hat{\phi} \in [0, 2\pi n_{\text{transit}})$, which we decompose as

$$\hat{\phi} = 2\pi k + \phi, \qquad k \in [0, n_{\text{transit}}] \tag{8}$$

where k is the toroidal transit number and $\phi \in [0, 2\pi)$ is the uniformly-spaced geometric toroidal angle within a single transit. The magnetic axis is first determined from a field-line trace by finding the fieldline starting position for which $R(\phi = 2\pi) = R(\phi = 0), Z(\phi = 2\pi) = Z(\phi = 0)$ using a root-finding algorithm. Then we map the flux surface of interest starting on the field line $\alpha = 0$, and using the magnetic axis location define the geometric poloidal angle

$$\hat{\theta}(\hat{\phi}, \alpha = 0) = \arctan(Z(\hat{\phi}) - Z_{\text{axis}}(\hat{\phi}), R(\hat{\phi}) - R_{\text{axis}}(\hat{\phi})), \quad \hat{\theta} \in [0, 2\pi\iota n_{\text{transit}}), \tag{9}$$

where the rotational transform ι is unknown. However, knowing $\hat{\theta}$ and $\hat{\phi}$ allows us to calculate the rotational transform on a flux surface,

$$\iota \simeq \frac{\Delta\theta}{\Delta\hat{\phi}}. \tag{10}$$

In practice, we trace the field line for $n_{\text{transit}} = 96 - 256$, with the optimal n_{transit} depending on ι and hence on how rapidly the field line samples the surface. Since the initial field line is chosen to have $\alpha_0 = 0$, its kth toroidal transit corresponds to $\alpha_k = 2\pi\iota k \pmod{2\pi}$. After sorting α, this gives us all the quantities of interest $R, Z, X, Y, \boldsymbol{B}$ in the ordered grid (α, ϕ) with a uniformly-spaced ϕ but a non-uniformly spaced α. It is also useful to additionally construct a Boozer-coordinate representation of the surface. We define the Boozer toroidal angle as

$$\phi_B = \phi + \nu, \tag{11}$$

where ν is the Boozer toroidal-angle shift. For the vacuum magnetic field generated by the coilset, ν is obtained directly along the initial field-line trace by integrating the streamline equation $d\phi_B/(\boldsymbol{B}\cdot\nabla\phi_B) = d\phi/(\boldsymbol{B}\cdot\nabla\phi)$ and using the fact that for a vacuum equilibrium in Boozer coordinates $\boldsymbol{B}\cdot\nabla\phi_B = B^2/G$. After simplification, the streamline equation becomes an ODE

$$\frac{d\nu}{d\zeta} = \frac{B^2}{G\,(\boldsymbol{B}\cdot\nabla\zeta)} - 1, \tag{12}$$

that can be integrated on a flux surface given a scalar G, the poloidal current in Boozer coordinates. Every quantity except G is known on the right side of (12). FTGen

¶ The idea of mapping a two-dimensional quantity on a surface to a set of field-aligned one-dimensional quantities was motivated by the ballooning transform

determines G by requiring the integrated shift ν to be bounded and centered over the trace,

$$\max_{\hat{\phi}} \nu(\hat{\phi}; G) + \min_{\hat{\phi}} \nu(\hat{\phi}; G) = 0. \tag{13}$$

By solving (13) as a root finding problem, we can calculate G and subsequently $\nu(\hat{\phi})$. Since these quantities are not periodic in ϕ for a given α, we fit 2D splines to the data in (α, ζ) coordinates using `scipy.interpolate.RectBivariateSpline`. Effectively, we have sampled every quantity on a single field line and mapped it as a two-dimensional surface quantity,

$$F(\hat{\phi}_{k,j}) \longrightarrow F(\alpha_k, \zeta_j), \quad F \in \{R, Z, X, Y, \boldsymbol{B}, \nu\}\,. \tag{14}$$

The first step is to use the definition $\alpha = \theta_{\text{PEST}} - \iota\phi$ and perform an affine transformation from (α, ϕ) to $(\theta_{\text{PEST}}, \phi)$. This allows us to calculate all the quantities on an ordered uniform PEST grid $(\theta_{\text{PEST}}, \phi)$ by evaluating $F(\theta_{\text{PEST}} - \iota\phi, \phi)$. Note that all the quantities are periodic in PEST coordinates. The second step is to evaluate all the quantities F on a uniform, ordered grid in Boozer coordinates (θ_B, ϕ_B). To do that, for each prescribed point (θ_B^*, ϕ_B^*) we use $\nu(\theta_{\text{PEST},\phi})$ to solve the root finding problem

$$\phi_B^* = \zeta + \nu(\theta_{\text{PEST}}, \zeta), \tag{15a}$$

$$\theta_B^* = \theta_{\text{PEST}} + \iota\nu(\theta_{\text{PEST}}, \zeta), \tag{15b}$$

giving the PEST coordinates on which we can then evaluate $F(\theta_{\text{PEST}} - \iota\phi, \phi)$, finally obtaining all the physical quantities on the prescribed Boozer grid points (ϕ_B^*, θ_B^*).

Thus, the FTGen algorithm gives us all the physical quantities F in three different coordinate systems (α, ϕ), $(\theta_{\text{PEST}}, \phi)$, and (θ_B, ϕ_B). This construction is useful for stellarator-mirror hybrid configurations because it allows us to parametrize a flux surface in multiple coordinate systems given a coilset. In particular, the method is different from the QFM surface approach, where one parametrizes a surface in double Fourier basis and solves for the coefficients along with the scalars ι and G, first developed by Giuliani. Our method is much more robust for SLAMs than the one developed by Giuliani.

The resulting surface data from FTGen are provided to the GVEC equilibrium fixed-boundary solver to calculate an ideal MHD equilibrium. The method is described in detail in the following section.

5. Calculating the Ideal MHD equilibrium using GVEC

In this section, we will use the shape of the flux surface Boozer coordinates are the appropriate choice for a SLAM

5.1. Neoclassical transport, MHD stability, and energetic particle transport in the unoptimized SLAM-A

6. Summary and conclusions

Need for integration optimization of the whole SLAM

Data availability Whoops!

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Appendix A. Stellartor mirror hybrids with $n_{\mathrm{FP}} > 2$

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